A comparison proof of rear-side reverse flow for the impulsive-cylinder Prandtl system

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Abstract

We give a direct comparison argument forcing reverse flow near the rear stagnation axis of a cylinder, provided the classical Prandtl solution persists sufficiently long. In the normalization with outer tangential velocity $2\sin\theta$, the explicit sufficient time is $T_{\mathrm{rev}} = 25/(9\pi) + 9/20 < 1.335$. The proof first bounds the rear-axis tangential derivative by a heat profile and then uses a polynomial supersolution on a fixed normal interval. It requires no initial trace for the tangential derivative: the heat comparison is started at a positive time and that time is subsequently sent to zero. The result is conditional on classical persistence and does not assert either that persistence or a van Dommelen–Shen collapse.

1 Setting and statement

The angular coordinate θ increases from the front point $\theta = 0$ to the rear point $\theta = \pi$ along the upper semicircle. The stretched outward normal coordinate is $R = (r-1)/\varepsilon$. Set

$$\mathcal{O} = (0,\infty)_R \times (0,\pi)_\theta, \qquad V(\theta) = 2\sin\theta.$$

The tangential and scaled normal velocities are denoted by $u(t,R,\theta)$ and $v(t,R,\theta)$. We consider

$$\begin{aligned} u_t + uu_\theta + vu_R - u_{RR} &= 4\sin\theta\cos\theta, \\ u_\theta + v_R = 0, \qquad v(t,R,\theta) &= -\int_0^R u_\theta(t,Z,\theta)\,dZ, \end{aligned} \tag{1.1}$$

with

$$u(t,0,\theta) = v(t,0,\theta) = 0, \qquad u(t,R,0) = u(t,R,\pi) = 0, \qquad u(t,R,\theta) \longrightarrow V(\theta) \tag{1.2}$$

as $R \to \infty$. The impulsive datum is

$$u(0,R,\theta) = V(\theta), \qquad R > 0. \tag{1.3}$$

It is incompatible with no slip at the initial-wall corner. In particular, no continuity of u at $t = R = 0$ is assumed.

The weak hypotheses used only to obtain the upper velocity bound are

$$\begin{aligned} u - V \in C([0,T']; L^2(\mathcal{O})), \qquad (u-V)|_{t=0} = 0, \\ u_R \in L^2((0,T')\times\mathcal{O}), \qquad k(t) := \|u_\theta(t)\|_{L^\infty(\mathcal{O})} \in L^1(0,T') \end{aligned} \tag{1.4}$$

for each $T' < T$. Boundary traces in (1.2) are understood in the appropriate sense. Notice that $u - V$, rather than u, is the square-integrable quantity on the half-line.

Theorem 1 (An explicit conditional reversal time)**.** *Let (u,v) solve (1.1)–(1.3) for $0 < t < T$, where*

$$T \geq T_{\mathrm{rev}} := \frac{25}{9\pi} + \frac{9}{20} \approx 1.334194. \tag{1.5}$$

Assume (1.4) *and enough positive-time regularity to differentiate the tangential equation once in* θ *and take its trace at* $\theta = \pi$*. For example, it suffices that* u, v *be smooth on every compact positive-time subset of the closed strip, with the stated boundary traces. Then there exist*

$$0 < t_r < T_{\mathrm{rev}}, \qquad R_r > 0, \qquad \pi/2 < \theta_r < \pi$$

such that

$$u(t_r, R_r, \theta_r) < 0. \tag{1.6}$$

More generally, the conclusion holds for any positive-time classical solution satisfying (1.1)–(1.2) *and the upper bound* $u \le V$*; for this formulation, neither* (1.3) *nor* (1.4) *is needed.*

The last formulation separates the actual reverse-flow argument from the comparison establishing $u \le V$. The proof of reversal uses no initial condition for u_θ, and does not differentiate the outer matching condition.

2 The upper velocity comparison

Lemma 2. *Under* (1.4)*, the boundary conditions and the regularity needed for the following weak comparison, one has*

$$u(t, R, \theta) \le V(\theta) = 2 \sin \theta. \tag{2.1}$$

Proof. Put $g = u - V$ and

$$\mathcal{L} = \partial_t + u\partial_\theta + v\partial_R - \partial_{RR}.$$

Since the forcing in (1.1) is VV_θ,

$$\mathcal{L}g + V_\theta g = 0. \tag{2.2}$$

At the wall $g = -V \le 0$, and at both angular endpoints $g = 0$. Also,

$$|v(t, R, \theta)| \le k(t)R. \tag{2.3}$$

Testing (2.2) with $g_+ = \max\{g, 0\}$, first with a cutoff in R, gives after removal of the cutoff

$$\frac{1}{2}\frac{d}{dt}\|g_+(t)\|_2^2 + \|\partial_R g_+(t)\|_2^2 \le 2\|g_+(t)\|_2^2. \tag{2.4}$$

The transport terms cancel because $u_\theta + v_R = 0$. There is no positive-part boundary contribution: $g_+ = 0$ at the wall and angular endpoints.

For completeness, choose a smooth cutoff $\chi_N(R) = \chi(R/N)$ equal to one on $[0, N]$ and zero on $[2N, \infty)$. The transport cutoff error is bounded by

$$Ck(t)\int_{\{R \ge N\}} |g(t)|^2 \, dR \, d\theta,$$

and the diffusion cutoff errors vanish using $g \in L_t^\infty L^2$ and $g_R \in L^2$. These errors tend to zero after time integration by dominated convergence and $k \in L^1$. Positive-time testing followed by passage of the starting time to zero is justified by the L^2 trace in (1.4). Thus (2.4) holds in its integrated form with zero initial positive part. Gronwall's inequality proves (2.1). □

3 The exact equation at the rear axis

Suppose, toward a contradiction, that there is no reverse flow on the rear half during the interval under consideration:

$$u(t, R, \theta) \geq 0 \qquad (R > 0,\ \pi/2 < \theta < \pi). \tag{3.1}$$

Define

$$w(t, R) := -u_\theta(t, R, \pi), \qquad W(t, R) := \int_0^R w(t, Z)\, dZ. \tag{3.2}$$

The letter W here denotes the normal velocity on the rear axis, not a tangential velocity or a vorticity. Indeed, incompressibility gives $v(t, R, \pi) = W(t, R)$.

Since $u(t, R, \pi) = 0$, taking the one-sided angular derivative of (3.1) and (2.1) gives

$$0 \leq w(t, R) \leq 2, \qquad 0 \leq W(t, R) \leq 2R. \tag{3.3}$$

To be explicit,

$$w(t, R) = \lim_{h \downarrow 0} \frac{u(t, R, \pi - h)}{h}, \qquad \lim_{h \downarrow 0} \frac{2 \sin(\pi - h)}{h} = 2.$$

Differentiate (1.1) in θ. On the rear axis, $u = 0$ and $u_R = 0$, so the terms $u u_{\theta\theta}$ and $v_\theta u_R$ vanish. If $a = u_\theta(t, R, \pi)$, the remaining equation is

$$a_t + a^2 + W a_R - a_{RR} = 4.$$

Consequently,

$$\boxed{w_t + W w_R - w_{RR} = w^2 - 4, \qquad W_R = w, \qquad w(t, 0) = 0.} \tag{3.4}$$

This calculation uses positive-time boundary regularity. No limit of u_θ as $t \downarrow 0$ or as $R \to \infty$ has been used.

4 A heat supersolution without an initial derivative trace

Lemma 3. *If (3.4) and (3.3) hold for $0 < t < T$, then*

$$w(t, R) \leq 2 \operatorname{erf}\left(\frac{R}{2\sqrt{t}}\right) \leq \frac{2R}{\sqrt{\pi t}}. \tag{4.1}$$

Proof. Fix $\eta > 0$ and compare on $t > \eta$ with

$$h_\eta(t, R) = 2 \operatorname{erf}\left(\frac{R}{2\sqrt{t - \eta}}\right), \qquad \operatorname{erf} z = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2}\, ds.$$

Then $h_{\eta,t} - h_{\eta,RR} = 0$, $0 \leq h_\eta \leq 2$, and $h_{\eta,R} \geq 0$. With $\mathcal{L}_W = \partial_t + W \partial_R - \partial_{RR}$,

$$\mathcal{L}_W h_\eta - (h_\eta^2 - 4) = W h_{\eta,R} + 4 - h_\eta^2 \geq 0. \tag{4.2}$$

Its wall value is zero and its initial value at each $R > 0$ is $2 \geq w(\eta, R)$. For $z = w - h_\eta$,

$$\mathcal{L}_W z - (w + h_\eta) z \leq 0. \tag{4.3}$$

Here z is bounded and $0 \leq w + h_\eta \leq 4$. The half-line comparison can be justified without prescribing a derivative limit at infinity. For $\delta > 0$, set

$$\Psi_\delta(t, R) = \delta e^{6(t - \eta)}(1 + R^2).$$

Using $W \geq 0$ and $w + h_\eta \leq 4$, one obtains

$$(\mathcal{L}_W - (w + h_\eta))\Psi_\delta \geq \delta e^{6(t-\eta)}(2R^2 + 2WR) \geq 0.$$

Thus $z - \Psi_\delta$ tends to $-\infty$ as $R \to \infty$ and has nonpositive parabolic boundary data. At the initial-wall corner, the required upper limit is nonpositive because w is continuous there at the positive time η, $w(\eta, 0) = 0$, and $h_\eta \geq 0$. The bounded-domain maximum principle, followed by removal of the distant boundary, gives $z \leq \Psi_\delta$. Letting $\delta \downarrow 0$ yields $w \leq h_\eta$.

Finally, send $\eta \downarrow 0$ at fixed t, R. This proves the first inequality in (4.1); the second follows from $e^{-s^2} \leq 1$. Only the comparison function was varied. The solution itself was never approximated. □

5 A polynomial barrier forcing reversal

Set

$$t_* := \frac{25}{9\pi}, \qquad d := \frac{6}{5}, \qquad A := \frac{14}{5}, \qquad s_* := \frac{9}{20}. \tag{5.1}$$

Since $2/\sqrt{\pi t_*} = d$, Lemma 3 gives, for $t \geq t_*$,

$$0 \leq w(t, R) \leq dR, \qquad 0 \leq W(t, R) \leq \frac{d}{2}R^2. \tag{5.2}$$

On $0 \leq R \leq 1$, put $s = t - t_*$ and define

$$B(s, R) = dR - AsR(1 - R), \qquad 0 \leq s \leq s_*. \tag{5.3}$$

This is an upper comparison function, and it is allowed to become negative. Write $S = As$, so $0 \leq S \leq 63/50$. Direct differentiation gives

$$\mathcal{L}_W B = -AR(1 - R) - 2S + W(d - S + 2SR). \tag{5.4}$$

We claim

$$\boxed{\mathcal{L}_W B - (w^2 - 4) \geq \frac{1}{250}.} \tag{5.5}$$

Indeed, (5.2) implies $w^2 \leq d^2R^2$. The polynomial

$$AR(1 - R) + d^2R^2$$

is increasing on $[0, 1]$, since its derivative is at least $2d^2 - A = 2/25$. Its maximum is therefore $d^2 = 36/25$. Also,

$$d - S + 2SR \geq -\frac{3}{50}, \qquad 0 \leq W \leq \frac{3}{5},$$

and consequently $W(d - S + 2SR) \geq -9/250$. Combining these bounds with $2S \leq 126/50$ gives

$$\mathcal{L}_W B - (w^2 - 4) \geq 4 - \frac{36}{25} - \frac{126}{50} - \frac{9}{250} = \frac{1}{250}.$$

The sign of B_R is not assumed positive in this calculation.

At the initial time $s = 0$, (5.2) gives $w(t_*, R) \leq B(0, R)$. At the normal endpoints,

$$w(t, 0) = B(s, 0) = 0, \qquad w(t, 1) \leq d = B(s, 1).$$

Subtracting the actual equation (3.4) from (5.5) and applying the usual linear parabolic comparison principle on $[0, 1]$ yields

$$w(t_* + s, R) \leq B(s, R). \tag{5.6}$$

For example, at the endpoint of the comparison interval,

$$B\left(\frac{9}{20}, \frac{1}{42}\right) = -\frac{1}{1400} < 0. \tag{5.7}$$

More generally, $B(s, R) < 0$ whenever

$$\frac{3}{7} < s \le \frac{9}{20}, \qquad 0 < R < 1 - \frac{d}{As}.$$

Thus the contradiction with $w \ge 0$ occurs strictly before $t_* + s_* = T_{\mathrm{rev}}$. This also covers a solution defined only for $t < T_{\mathrm{rev}}$.

Completion of the proof of Theorem 1. Lemma 2 provides $u \le V$. If no rear-side reverse flow occurred before T_{rev}, the angular endpoint inequalities would give (3.3). The heat and polynomial comparisons would then contradict its lower bound $w \ge 0$. Hence (1.6) holds at some time strictly before T_{rev}.

Equivalently, if one formulates the contradiction solely as $w \ge 0$ throughout the interval, its failure gives $u_\theta(t_r, R_r, \pi) > 0$ at some time and normal position. Positive-time differentiability then gives

$$u(t_r, R_r, \pi - h) = -h\, u_\theta(t_r, R_r, \pi) + o(h) < 0$$

for sufficiently small $h > 0$. This locates the reverse flow on the rear half. □

6 Scope of the conclusion

The comparison constant T_{rev} is explicit and is not claimed to be optimal. All times above use the normalization $V(\theta) = 2\sin\theta$. The proof applies to a solution that exists with the stated positive-time regularity for a sufficiently long interval. It does not prove a common lifespan for approximating solutions, and it does not require such approximations.

The logical conclusion for the impulsive cylinder is therefore: a classical solution satisfying the comparison hypotheses cannot remain free of rear-side reverse flow throughout $(0, T_{\mathrm{rev}})$. If classical existence breaks down earlier, this argument alone does not decide whether reverse flow has already occurred. Finally, reverse flow is the strict sign change (1.6); it is not equivalent to fixed-label van Dommelen–Shen collapse of the tangential flow map.